Hassan Kamal

Senior Data Architect | Healthcare, Regulated Data & AI
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Senior Data Architect with direct medicines-regulation and public-health experience delivering data strategy and modelling for Licensing, Imports and Product Regulation. Combines conceptual-to-physical modelling, secure integration and governed cloud data platforms with strong controls for source lineage, provenance, quality and auditability. Has designed data architectures supporting AI-enabled risk analysis and predictive analytics, alongside NHS Digital/NHSX-aligned requirements. Brings hands-on database engineering, architecture assurance and effective collaboration across business, engineering and governance teams.

Core Skills

- **Healthcare & Regulatory Data** - MHRA medicines licensing, imports and product-regulation domains; public-health requirements aligned to NHS Digital/NHSX; FHIR familiarity.
- **Data Modelling & Semantics** - Conceptual, logical and physical models; business rules, entities and relationships; canonical schemas; reference/master data; relational and dimensional modelling.
- **Provenance, Quality & Governance** - Source-to-consumption lineage, metadata, traceability, stewardship, data quality, audit-ready reporting, access controls and architecture assurance.
- **Data Platforms & Integration** - AWS, Azure, Databricks, Fabric, Synapse, Snowflake and data lakes; SQL; APIs, ETL/ELT, Kafka, batch, streaming and event-driven integration.
- **AI/ML Data Foundations** - Governed inputs and reusable data products for prediction and risk analysis; AI-service integration; quality controls and consistency of schemas and interfaces.
- **Engineering & Collaboration** - Python, SQL Server, Oracle and PostgreSQL; Erwin, PowerDesigner, Sparx EA, BiZZdesign and Collibra; multidisciplinary delivery and design authorities.

Work Experience

Correla / Xoserve **- May 2026 - present**
Senior Data Architect

- Lead data architecture and governance across Xoserve's regulated managed-service estate, aligning platform, supplier and integration designs to business and data-control requirements.
- Defined a 13-week evidence-led assessment against Ofgem's 11 Data Best Practice principles, identifying data ownership, lineage, quality and governance gaps and a phased improvement plan.
- Shape target-state models and integration architecture across SAP IS-U/BW, Azure/AWS, SaaS products and APIs, mapping source dependencies and controls for downstream consumers.
- Assess raw-source versus curated access to the External Data Portal, addressing historical reconstruction, reconciliation, security and the traceability of published information.
- Define metadata, stewardship and data-quality expectations; challenge supplier designs and support consistent architecture decisions and change governance.

London North Eastern Railway (LNER) **- December 2025 - May 2026**
Lead Data Architect

- Defined target-state and transition architecture for the AWS Intelligent Data Platform, connecting source systems to governed lake and curated analytics layers.
- Owned conceptual, logical and physical models across performance, operations and assets, including temporal integrity and standardised business definitions.
- Designed batch, streaming and event-driven integrations across Network Rail and internal sources, specifying freshness, resilience and data-consumption requirements.
- Designed trusted source-to-consumption data products for three predictive AI/analytics use-case families: performance prediction, delay attribution and asset availability.
- Established metadata, lineage, ownership and data-quality foundations for analytics consumers, working with engineers, analysts and operational stakeholders.

Balfour Beatty **- March - December 2025**
Lead Data Architect - M25 Programme

- Led data architecture for a £2m programme across finance, operations and asset management, coordinating enterprise stakeholders and design decisions.
- Structured raw-to-curated transformations and lineage for executive reporting, reducing decision latency from weeks to days.
- Reduced reporting discrepancies by 30% through consistent business definitions, glossary, reference data and quality controls.

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- Chaired the Data Council and Technical Design Authority, providing governance for standards, ownership, design assurance and exceptions.
- Designed a domain-oriented Azure/AWS architecture and governed ingestion of native Esri GIS data.

Insurwave ~ June ~ Nov 2024
AI Platform ~ Senior Data Architect
Insurtech platform delivering real-time, AI-enabled risk-exposure mapping for insurers, brokers and clients.

- Designed data and solution architecture for an AI-enabled risk-exposure platform, connecting structured property and risk-zone data models to AI services.
- Standardised canonical entities and integration semantics using ACORD across 10+ counterparties, improving data consistency at broker/insurer boundaries.
- Defined API boundaries and access controls; secured CTO/CEO support for a 36-month scale-up roadmap.

Shell ~ May 2022 ~ May 2024
Senior Data Architect ~ Chemicals & Products

- Defined an Azure lakehouse architecture and transition roadmap consolidating 12+ legacy stores into governed data products.
- Implemented Data Vault 2.0 and MDM standards, improving reporting accuracy by 25% and reducing reconciliation cycles by 35%.
- Architected SAP, Salesforce and Azure integration across 15 global markets and mentored 12 architects and modellers.

Medicines and Healthcare products Regulatory Agency ~ Feb 2020 ~ April 2022
Senior Data Architect ~ Transformation

- Architected cross-agency public-health data strategy and data-platform designs during pandemic response, aligning regulatory requirements with NHS Digital/NHSX.
- Modelled data across medicines Licensing, Imports and Product Regulation, shaping the structure of a regulatory data hub and secure transition from legacy sources.
- Embedded master-data management, traceability and quality controls to strengthen source provenance, audit-ready reporting and regulatory transparency to Parliament.

HSBC ~ Feb ~ Dec 2019 | London (Hybrid)
Data Architect ~ Transformation

- Led IFRS 16 data architecture across Finance and Risk, defining source mappings, target data structures and traceable integration controls.
- Designed governed automated pipelines, cutting manual processing by 70%; applied BIAN and migrated legacy EUCs towards GCP/Azure solutions.

Insight Investment ~ Dec 2017 ~ Dec 2018 | London (Onsite)
Data Modeler ~ ThinkFolio

- Modelled investment workflows for the ThinkFolio platform; automated MiFID reporting in SQL/VBA, saving 10 hours per week.

Earlier Financial Services and Engineering Experience
JP Morgan, BlackRock, HSBC, ABN Amro, Deutsche Bank, Barclays | 2000–2016
Progressed from database and data-warehouse engineering into modelling and architecture for risk, trading, derivatives and regulatory reporting. Designed production schemas, ETL, migrations and integrations with a strong focus on SQL performance and data reliability.

Education and Professional Development

BSc Computer Science; University of Westminster. Equity Derivatives training; JP Morgan. TOGAF certification in progress; working knowledge of DAMA-DMBOK and CDMP Foundation material.

Additional Interests and Responsibilities

PADI Rescue Diver; RYA Day Skipper; woodworking and skiing. Former director of the management board for a 150-apartment residential development.